

Basics

Name Seongwon Kim

Label Integrated MS/PhD Student

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Summary Integrated MS/PhD student in Mechanical and Robotics Engineering at GIST, advised by Prof. Pyojin Kim. Research interests include Computer Vision, Robot Vision, and SLAM, with a focus on LiDAR/visual-inertial odometry, 3D scene reconstruction, and robust state estimation.

Work

2026.03 - PRESENT

Graduate Research Assistant

Machine Perception and Intelligence Lab (MPIL), GIST

Conducting research on Computer Vision, Robot Vision, and SLAM under the supervision of Prof. Pyojin Kim.

- LiDAR/Visual-Inertial Odometry
- 3D Scene Reconstruction
- Robust State Estimation

Education

2026.03 - PRESENT

📍 Gwangju, South Korea

Integrated MS/PhD

Gwangju Institute of Science and Technology (GIST)

Mechanical and Robotics Engineering

- Computer Vision
- Robot Vision
- SLAM

2025.06 - 2025.08

📍 Berkeley, CA, USA

Visiting Student

University of California, Berkeley

Summer Session

2022.03 - 2026.02

📍 Gwangju, South Korea

B.S.

Gwangju Institute of Science and Technology (GIST)

Electrical Engineering and Computer Science

Awards

2026.10

Best Oral Presentation Award

IEEE/RSJ IROS 2026 Student Research Workshop

For "Bridging the Scale Gap: Cross-View Localization from Dense Rover LiDAR to Coarse Lunar DEMs" (selected for oral presentation).

Publications

2026.10

Bridging the Scale Gap: Cross-View Localization from Dense Rover LiDAR to Coarse Lunar DEMs

IEEE/RSJ IROS 2026 Student Research Workshop

Seongwon Kim, Minseok Song*, Seonmo Yang, Soumyadeep Chatterjee, Ryan Soussan, Seokju Leet, Pyojin Kim† (*equal contribution, †corresponding). Selected for oral presentation; Best Oral Presentation Award.*

Skills

🔧 Programming

Python

C++

🔧 Frameworks & Tools

ROS / ROS2

PyTorch

🔍 Research Areas

Computer Vision

Robot Vision

Simultaneous Localization and Mapping (SLAM)

LiDAR / Visual-Inertial Odometry

3D Scene Reconstruction

Languages

Korean

Native speaker

English

Professional working proficiency

Interests

Computer Vision & Robot Vision

Visual perception for autonomous systems

Depth estimation and 3D reconstruction

SLAM & State Estimation

LiDAR-Visual odometry

Loop closure and global localization

Robust estimation under challenging conditions